

BGIN Meeting Report

Date: TBD

Location: Virtual Meeting

Executive Summary

The meeting focused on the post-quantum cryptography (PQC) migration for blockchain technologies, particularly Bitcoin and Ethereum. The discussion included an overview of the challenges in PQC migration, a presentation by Kigeneegan on the technical difficulties and stakeholder incentives, and details about a Japanese government competition to accelerate this process. The competition aims to develop standardized solutions for PQC migration and key management protocols, with a focus on creating a global testbed and ensuring open access to the winning solution.

Key Discussion Points

- **PQC Migration Challenges:** Kigeneegan highlighted that while the necessity for PQC migration is widely recognized, blockchain's decentralized nature makes this process particularly challenging. Issues include the lack of centralized administration, long-term data validity, and potential impacts on asset prices.
- **Stakeholder Incentives:** The discussion covered how different stakeholders (end users, miners, investors, node operators) have conflicting incentives that complicate reaching a consensus on migration strategies.
- **Technical Difficulties:** PQC algorithms generally perform less efficiently than traditional ones, leading to larger key and signature sizes. This can affect blockchain performance and decentralization.
- **Migration Strategies:** Various strategies were proposed, including zero-knowledge proofs, hybrid signatures, and pure PQC migration. Each strategy has its pros and cons in terms of burden on users, operational overhead, and backward compatibility, creating a trilemma.
- **Japanese Government Competition:** The competition will run from 2026 to 2028, focusing on PQC migration and key management protocols. It aims to create a global testbed and ensure the winning solution is open source and globally standardized.

Action Items and Next Steps

- **Competition Specification and Criteria:** The Japanese government will define the competition's scope and evaluation criteria in late 2026, with input from academic workshops.
- **Global Testbed Development:** Begin will create a global testbed with nodes distributed across North America, Asia, Australia, and Europe to evaluate PQC solutions.

- **Stakeholder Engagement:** Further dialogue among blockchain developers, researchers, and stakeholders is necessary to align on migration strategies and criteria.
- **Community Participation:** Encourage participation in the competition from the global community, including quantum researchers and blockchain experts.

Detailed Session Summary

The meeting began with a brief introduction and then transitioned into a detailed discussion on post-quantum cryptography (PQC) migration for blockchain technologies. Kigeneegan Fukuda from Virginia Tech presented the challenges and complexities involved in PQC migration for Bitcoin and Ethereum.

Technical Challenges

Kigeeegann explained that current cryptographic primitives used in blockchains are vulnerable to quantum attacks, which could lead to unauthorized asset transfers. The decentralized nature of blockchains complicates the migration process due to the lack of a central authority and the need for long-term data validity. He also discussed how PQC algorithms generally have larger key and signature sizes, affecting blockchain performance and decentralization.

Stakeholder Incentives

The presentation covered the various incentives of different stakeholders: - **End Users:** Minimize economic and operational burden. - **Blockchain Ideologists:** Preserve decentralization. - **Miners:** Maintain profitability and maximize transaction fees. - **Investors:** Increase asset prices. - **Node Operators:** Avoid increased computational and storage overhead.

Migration Strategies

Kigeneegan introducedproposed several migration strategies: - **Zero-Knowledge Proofs:** Use preimage of the signing key to ensure quantum security. - **Hybrid Signatures:** Combine PQC and traditional ECDSA keys for backward compatibility. - **Pure PQC Migration:** Generate new PQC keys and move assets from classic addresses. - **Centralized Solutions:** Use quantum-resistant proxies or emergency rollbacks.

Japanese Government Competition

The meeting then shifted to the announcement of a competition by the Japanese government, aimed at accelerating PQC migration. The competition will run from 2026 to 2028 and focus on developing standardized solutions for PQC migration and key management protocols. Key points include: - **Scope and Criteria:** Defined in late 2026 with input from academic workshops. - **Global Testbed:** Creation of a testbed with nodes distributed globally. - **Open Access:** The winning solution will be open source and free to use.

Q&A Session

The discussion included questions on the testing environment, intellectual property (IP) policies, and the differences in migration approaches between Bitcoin and Ethereum. It was noted that Ethereum's embedded logic and flexibility might make it easier to retrofit PQC solutions, while Bitcoin's simpler value-based approach could lead to different strategies.

Also, the audience brought up abandoning legacy chains in favor of the quantum-ready chain idea, while the other highlighted the difficulties in preserving asset value.

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Conclusion

The meeting concluded with a call for further stakeholder engagement and participation in the competition. The goal is to develop robust, standardized solutions for PQC migration that can be widely adopted across the blockchain industry.

Appendix: Full Meeting Transcript (Anonymized)

22.7 --> 26.7: So, I'm not sure how much time I get, so I'm going to give it a while to get going.
26.7 --> 32.4: So, let's see, I'm going to go to the airport.
33.5 --> 35.2: I thought my nose closed for a second.
63.5 --> 65.2: Oh, no, no, no.
74.1 --> 74.7: Okay.
75.0 --> 78.8: We're going to continue the workshop, if that's okay, everyone.
84.8 --> 86.6: I know.
101.8 --> 105.5: Yeah, I think you'll have a lot of email IDs from the last week.
105.5 --> 108.4: Yeah, yeah, yeah.
112.0 --> 121.2: Yeah, yeah, yeah.
121.2 --> 134.8: Yeah, yeah, yeah.
134.8 --> 135.1: Yeah.
149.8 --> 154.5: So, I can see that we've got Keegan online.
155.0 --> 156.7: Peace out.
158.4 --> 163.7: So, I'm hoping that Keegan can help me with this, because he's more on the post-quantum cryptography side.
164.3 --> 175.5: But I will chair this workshop, and we have half an hour to discuss the PQC, crypto agility, and signature migration of our keys.
176.1 --> 184.3: And that is then going to, I'm just going to quickly flick through to give everyone in the room, like, a bit of a high level of where we're going.
184.3 --> 187.8: The final conversation is beginning a common lexicon.
188.3 --> 194.8: And this is most relevant to the regulatory pieces and the threats and harms we experience in the industry.
195.0 --> 201.1: So, the different hacks and the different sort of manipulations of smart contracts, et cetera.
201.1 --> 211.2: What Begin is trying to do is build a common, like, be a cu

stodian of a machine readable database for threats and harms in the space.

211.5 --> 228.2: And is coordinating with MITRE, NIST, and SEAL 911, which is the crypto social community initiative that supports people who have been hacked or exploited, exchanges, et cetera.

229.2 --> 240.4: So, we do have, we have an hour from here, and we'll start with the PQC, and then we'll move on to the common lexicon.

241.2 --> 247.2: So, this is a reflection on some of the announcements that were made last week in Tokyo.

247.2 --> 259.6: And I'm going to quickly pass it to either Keegan or Professor Matsuura to provide the high level timeline of the PQC migration.

260.2 --> 263.3: We're focusing specifically on Bitcoin and Ethereum.

264.2 --> 265.5: I want to make that clear.

266.3 --> 273.0: These are probably the most, they have the most assets, so they're the most relevant to make quantum secure.

273.6 --> 275.8: And that's the nature of the work.

275.8 --> 281.6: Most of this work is also relevant to other networks, but it would have to be repurposed.

283.3 --> 299.6: So, I might pass to Keegan to quickly explain the timeline and then what we're doing with the Japan government and NIST as Begin initiating this competition.

299.6 --> 305.9: And if Professor Matsuura is still there, I know he had another meeting to go to, but Keegan, if you could.

306.3 --> 312.5: So, before starting, so, you know, this is just a 30 minute session.

313.4 --> 315.9: And, you know, so that's right.

316.0 --> 324.5: So, what I'm showing is about their current, you know, plan by NIST or other government authorities in the other countries.

325.7 --> 334.2: I don't know that this, you know, time, you know, this timeline applied to the blockchain, you know, PQC migration for blockchain.

335.4 --> 342.5: But, so at this session, so, you know, I would like to ask Keegan to provide us some 10 minutes or 15 minute presentation.

342.7 --> 349.1: What is there, what the true problem regarding the PQC migration for blockchain technology.

349.1 --> 355.4: It's a kind of the different types of problem there in that blockchain world.

356.3 --> 360.1: So, we have that common understanding that what is the problem for blockchain first.

360.7 --> 370.0: So, I would like to provide five minutes, a brief announcement about that upcoming initiative by Japanese government.

370.0 --> 375.7: Then, so we have, we would like to hear that opinion from the audience in the room about, you know,

375.8 --> 380.2: and I would like to encourage that participation to that initiative by Japanese government.

380.5 --> 387.6: Anyway, I would like to pass the microphone to Keegan to provide, to have the common understanding of the problem for the blockchain world.

389.6 --> 392.7: His presentation will be in 10 minutes or 15 minutes presentation.

394.0 --> 395.7: Keegan, did you want to share your screen?

406.0 --> 408.3: We can't hear you.

408.3 --> 424.9: Can you hear me?
429.1 --> 430.0: Yeah, that's better.
431.7 --> 432.4: Can you hear me?
433.9 --> 435.9: Yeah, we can hear you now.
449.5 --> 450.0: Hello?
451.2 --> 451.5: Hello?
451.8 --> 452.3: Can you hear me?
452.9 --> 453.1: Yep.
453.8 --> 454.9: Okay, finally.
455.4 --> 455.8: Sorry about that.
456.4 --> 458.2: So, let me share the screen.
460.6 --> 462.2: Is it going through?
463.3 --> 463.9: Okay.
465.5 --> 466.1: Hello.
466.4 --> 471.1: So, I'd like to start the presentation like for 10 minutes, like, yeah,
471.2 --> 475.2: structuralizing the difficulty of the blockchain PQC transition.
475.7 --> 477.7: And this is Keegan Fukuda from Virginia Tech.
477.7 --> 485.4: And so, I mean, as you guys know, the cryptographic primitives, you know,
485.4 --> 493.1: foundational to blockchain are not threatened by the quantum computer, which I call CRQC in this presentation.
494.1 --> 500.2: And so the quantum attacker may derive the signing key from the verification key and they can force the valid signature.
500.2 --> 512.0: And they can execute the unauthorized asset transfers, like not your key, not your coins, that, you know, core principle of blockchain never stands anymore after the Q day happens.
513.0 --> 523.1: And so while the necessity for the PQC migration is widely acknowledged, the blockchain remains an extremely challenging domain for the such migration.
523.1 --> 534.7: And this is because firstly, there is no centralized administrator, the catastrophic impact on the compromise, also like off-chain data may require like extremely long term validity.
535.2 --> 538.1: So, this is quite different from the other, you know, conventional systems.
538.4 --> 547.6: And also, because this is the financial system anyway, the PQC migration will eventually affect on the asset price.
547.6 --> 563.4: And there, you know, there's some conflict of interest, like the misaligned incentives and, you know, conflicting speculation among stakeholders are kind of actually making the, you know, reaching the consensus difficult.
564.4 --> 571.3: And so this research is done by me like last year.
571.8 --> 581.7: And well, basically, because, you know, the discussion about the PQC migration within academia or blockchain community is really getting fragmented right now.
582.0 --> 588.4: And also, there's no comprehensive framework, you know, the technical catalogs, you know, technical and governance challenges.
588.4 --> 600.5: So I went through like all the discussions happening in the blockchain community and academia and list the difficulties from both, you know

now, technical aspect and also operational aspect.

601.3 --> 608.7: And this project has began in 2025 at the NISPO Cryptology Workshop.

608.7 --> 614.4: So I did some kind of problem presentation in this workshop.

615.0 --> 621.4: And this is chosen for 10 slots for presentations by NISPO.

621.5 --> 631.0: So that showcases like how NISPO is also caring about the blockchain as the, you know, PQC migration, like edge cases, I'll say.

631.0 --> 642.3: And as I said, all the difficulty spans, like, I mean, range from the, for example, in applicability of the conventional approach.

642.7 --> 646.6: Also, even the updating the decentralized network is kind of hard.

647.3 --> 654.1: But in this presentation, I'll just go through the technical difficulties first.

654.5 --> 660.3: And also, after that, I'll cover the incentive analysis very quickly.

661.3 --> 676.7: So first of all, the technical side, as you guys may know, the PQC algorithm has much lower performance compared to the classic ECDSA or RSA.

677.3 --> 684.0: Like ECDSA or Bitcoin, they have verification key for 33 bytes and signature 77 bytes.

684.0 --> 692.2: But the Falcon, for example, the one, the PQC signature scheme currently standardized by NIST.

692.6 --> 704.0: They have the verification key for 897 bytes, signature 66 bytes, which is much bigger than the current one, obviously.

704.5 --> 708.1: And there is a tradeoff between the throughput and the decentralization.

709.6 --> 719.8: You know, I mean, the increase in the block size will eventually damage the decentralization of the blockchain.

720.3 --> 722.4: So this is already a big problem.

723.2 --> 728.9: And so even some migration, some mitigation plans are proposed for this problem.

729.1 --> 734.4: But like this tradeoff is the kind of fundamental and difficult to solve anyway.

734.4 --> 755.8: And also, even other, I mean, other than the transaction signing algorithm, there should be some migration happen for the other primitives such as signature aggregation or other polynomial commitment, which is widely used in Ethereum.

755.8 --> 766.5: Also like scaling solution like layer 2, hash time lock contract, also smart contracts on Ethereum or other chains.

767.0 --> 771.1: That is also a problem because we cannot update the smart contract code.

772.2 --> 781.8: And these like other primitives or other tech stack that requires some migration really depends on each chain instance.

781.8 --> 787.9: So like the rigorous analysis on each chain should be done before the PQC migration.

788.6 --> 801.2: And to talk about the incentive side, as I said, the incentive doesn't align each other among the PQC migration.

801.7 --> 808.9: And here I list very simplified stakeholder incentives and their intention on focusing on Bitcoin.

808.9 --> 818.5: But what important is, for example, in end user, they want

to minimize the economic and operational burden within the migration.

819.2 --> 823.5: And the blockchain ideologies may want to preserve the decentralization.

824.2 --> 827.7: And miners, obviously, they want to maintain the profitability of the mining.

828.0 --> 831.6: And also they want to maximize the total transaction fee revenue.

832.3 --> 836.3: And there's investors who want to increase the price of their holding.

836.3 --> 846.8: And also there are node operators who don't want to have any overhead, who want to avoid increasing operational overhead,

847.3 --> 852.5: such as computational, higher computational resource or like higher storage.

853.2 --> 857.6: And also this is the shared interest among all stakeholders.

857.9 --> 865.8: We want to avoid hard forks anyway, to ensure the robust security and also preserve the overall value of the asset.

866.8 --> 872.1: So let's see the proposed TX migration plan very briefly.

872.6 --> 881.6: So one solution, I mean, the one migration plan proposed by the community is using the classic secret.

882.1 --> 885.7: We don't want to even use the PQC signature scheme in this case.

886.1 --> 892.0: So we do the zero-knowledge proof of the mnemonic code, which is the preimage of the signing key.

892.4 --> 894.2: And this remains quantum secure anyway.

894.2 --> 902.0: So if we do that, this will ensure, I mean, this will enable the short term migration period.

902.4 --> 906.6: Also, we can keep using the same, not a key, but the same secret information.

907.3 --> 912.3: Also, there is the other plan called hybrid signatures or account abstraction.

913.1 --> 921.4: This is like we do generate the post-quantum signature signing key, but we can also keep using the ECDSA key at the same time.

921.4 --> 924.5: So we can ensure the backward compatibility in this case.

925.3 --> 927.9: And this will ensure the end users less burden.

929.0 --> 938.7: But node operators, developers are going to have more cost in operation because these are causing the overhead.

939.7 --> 944.5: And backward compatibility is provided to avoid the hard fork.

945.0 --> 949.8: And also miners will gain no profit because there is no transaction happening in the migration.

949.8 --> 955.1: And the other is pure PQC migration.

955.4 --> 962.7: So users just generate the PQC key and move their assets from the classic address to the PQC address.

963.3 --> 968.8: So end users are going to have more burden and node operators, developers having less cost.

969.4 --> 971.1: And there's no backward compatibility.

972.1 --> 977.9: So it's going to have some probability of the fork and also miners can get the profit for this.

977.9 --> 987.5: And the other is, well, why don't we rely on some centralized implementation, for example,

987.9 --> 991.8: like quantum resistant proxy as a vault for the migration.
992.3 --> 1001.2: Also, some other plan is proposing emergency rollback after we find out the quantum computers attack.
1001.2 --> 1009.5: And this will, this kind of approach will ensure the decentralized, I'm sorry, like shorter migration.
1009.7 --> 1014.5: But this will also damage the decentralization by making the single point of failure.
1014.9 --> 1017.3: And it's going to have less overhead.
1017.3 --> 1023.0: Like by concatenating all these proposal migration plans,
1023.0 --> 1030.3: we kind of see the Trelema-ish relationship between these migration plans.
1031.7 --> 1037.7: And Trelema between decentralization and the security and overhead.
1038.3 --> 1043.2: So besides the technical problem, as shown in this Trelema,
1044.3 --> 1049.5: there is no silver bullet in the PQC migration, in the operational side, especially.
1049.5 --> 1058.2: And this kind of relationship even can be applied for a specific asset type, which is called dormant assets.
1058.9 --> 1060.4: But I'm going to skip this.
1060.9 --> 1066.4: And anyway, so what's missing is like, we don't even know the PQC migration requirements.
1066.9 --> 1070.2: And also the definition of the PQC secure blockchain.
1070.5 --> 1073.1: What's a migration completion for a blockchain?
1073.6 --> 1076.8: I mean, we don't know this, such like definitions.
1077.4 --> 1080.6: And also the defining the migration scope is important.
1081.0 --> 1088.7: As I covered the first step, the migration doesn't really, it's not limited to the transact signature scheme.
1089.0 --> 1096.2: But we have much more, you know, I mean, the blockchain protocol is getting more complex and involving more stakeholders
1096.2 --> 1100.5: and involving more technologies like, you know, commitment, like scaling, whatever.
1101.0 --> 1103.8: And we have to define the scope for this.
1104.8 --> 1111.5: And also, like, you know, designing the balance, then the incentive align migration strategies is also important.
1112.3 --> 1121.0: And as Dr. Matsuo will continue, we will have the tech competition for this PQC migration.
1121.7 --> 1128.8: And what's the evaluation criteria given the presentation, given the, you know, this difficulty and complex current situation.
1129.1 --> 1141.2: I mean, defining the scope criteria for such competition is also, it's going to be, I mean, the next topic we're going to discuss in the next 20 minutes.
1141.8 --> 1144.5: And this concludes my kickoff slide.
1144.7 --> 1145.7: And thank you for listening.
1146.7 --> 1148.9: Thank you, Keegan.
1149.5 --> 1153.8: I guess if we could move on to the competition announcement side.
1162.0 --> 1166.5: Okay, so let me explain a little bit more about that competition.
1167.5 --> 1174.1: It's like the competition, you know, generally held by NI

ST.

1174.5 --> 1185.0: NIST holds cryptographic algorithms competition like, you know, AES or SHA-3 or post-quantum cryptography so far.

1186.0 --> 1189.2: So they're using that very academic process.

1189.6 --> 1200.5: But it takes, so it spends five years, seven years to pick up that, you know, certified standardized cryptographic primitive.

1201.7 --> 1205.3: And it's for this competition by Japanese government.

1205.3 --> 1210.5: So there are the two subtopics.

1210.6 --> 1211.7: One is the PQC migration.

1211.9 --> 1217.3: The other is a key management protocols for the crypto as set custody.

1218.6 --> 1222.2: But so we are, I'm not focusing on that discussion about PQC migration.

1222.9 --> 1228.3: So this is a three years competition from 2026 to 2028.

1229.3 --> 1244.6: In 2026, so, you know, Japanese government will decide the specification of the competition itself, what kind of technology.

1245.1 --> 1251.2: So Japanese government will choose or what kind of evaluation criteria.

1251.5 --> 1253.8: So they will apply to the competition.

1253.8 --> 1260.1: So they will hold their academic workshop in October, in late October in Japan.

1261.0 --> 1271.5: So they will make a call for papers about the specification of the competition and the evaluation criteria from academic point of view.

1271.8 --> 1273.5: And the 2027 is a year one.

1273.9 --> 1276.3: So I'm sorry that this is a typo.

1276.8 --> 1278.9: In 2027, the second year.

1278.9 --> 1292.0: So the applicant will provide some documents about the technology itself and the implementation by software code and the self-evaluation document.

1292.2 --> 1296.2: So it is very similar to that NIST competition application document.

1297.1 --> 1302.2: And the beginning will be a part of that evaluation committee.

1302.6 --> 1305.3: We will start the evaluation of the whole application.

1306.3 --> 1310.1: So we will begin to create a global testbed.

1310.4 --> 1318.5: You know, three nodes in North America, three nodes in Asia, two nodes in Australia, something like that.

1319.1 --> 1322.6: So begin we will create a global scale testbed.

1322.9 --> 1326.4: It contains 10 to 30 nodes in the world.

1327.3 --> 1331.1: So do that implementation evaluation.

1332.3 --> 1341.9: So in the year three, 2028, the Japanese government and the evaluation committee select the winner of the competition.

1342.8 --> 1349.7: The major difference against the NIST competition is that the NIST competition will not provide any prize.

1350.3 --> 1354.3: But the Japanese government will provide tons of money to the winner.

1354.3 --> 1362.6: The total amount of prize to all competition will be a million dollars.

1363.6 --> 1368.7: Anyway, so the Japanese government will organize a three-year competition.

1369.0 --> 1371.5: It's a global competition, so anyone in the world can apply.

1372.2 --> 1383.0: Then after the selection of the winner, the Japanese government will provide some support to make it a global standard, like a BGN standard, ISO standard.

1384.0 --> 1387.9: This is what the competition looks like.

1392.0 --> 1394.3: We have 10 minutes left for the discussion.

1396.3 --> 1401.8: It could be the topic of discussion during these 10 minutes.

1402.2 --> 1404.8: What kind of technology do we need?

1406.2 --> 1408.9: What kind of technology should be evaluated?

1409.3 --> 1410.8: What are the evaluation criteria?

1411.8 --> 1416.0: Anyway, this is my short explanation about the competition.

1417.7 --> 1418.1: Thank you.

1420.8 --> 1427.5: I'm not sure how many in the room are aware of the quantum threat cryptography.

1428.5 --> 1440.8: But this initiative is to add some game theory to accelerating the PQC migration for our Bitcoin and Ethereum blockchain networks.

1441.4 --> 1455.9: So I guess the question that I would pose to the room right now is, which aspects of the competition do you think would be most relevant to maybe your work?

1456.0 --> 1468.3: Or if you want any further clarity on what PQC cryptography actually really means, we've got Keegan here to provide a bit more color and insight on that.

1480.0 --> 1485.0: There's already a number of post quantum cryptography solutions out there.

1485.3 --> 1489.1: How are they deficient in addressing this problem?

1490.0 --> 1495.0: In other words, is some sort of market research already being conducted on what's available?

1496.5 --> 1499.0: Yes, so that's a really good question.

1499.0 --> 1508.8: And this competition in particular is to use that research in the blockchain industry and space.

1509.4 --> 1514.2: So the blockchain has unique challenges with PQC.

1515.0 --> 1520.3: And those are not being fully addressed by the existing research efforts.

1521.1 --> 1528.6: So we're not looking for, and this was discussed in Tokyo, we're not looking for people to invent new PQC signature schemes.

1528.9 --> 1532.2: There are plenty that work in the PQC world.

1532.5 --> 1542.7: We're looking for those who use those signature schemes for Bitcoin or for Ethereum in either the transaction side or in the signature side.

1543.8 --> 1548.3: They're the two sort of resistant technologies that need to emerge.

1549.3 --> 1553.1: So yeah, that's how I would answer that question.

1553.3 --> 1558.8: This adds that specific frame within the blockchain industry.

1559.8 --> 1567.8: Because part of our purpose at Begin is to facilitate this multi-stakeholder dialogue.

1568.7 --> 1576.1: And part of the, it's both the benefit but also the diffi

culty with the crypto blockchain industry.

1576.1 --> 1582.1: It is decentralized and fragmented and Ethereum's working on its thing over here.

1582.2 --> 1584.8: And the Bitcoin core developers are working on its thing over there.

1585.0 --> 1588.0: And they sometimes talk to each other, but they don't really talk to each other.

1589.0 --> 1597.9: So developing some sort of standard pathway for those coming up with the solutions to these problems is what we're trying to facilitate.

1597.9 --> 1605.1: And we're hoping that the competition adds that oomph to that communication beginning.

1606.8 --> 1615.2: I could just be the IP lawyer in me, but I see people really want to give up their secret sauce.

1616.4 --> 1624.4: Is the intention to have the solution, if it is derived, to be open access and universally available?

1624.4 --> 1628.3: Yeah, it has to be because it's a public permissionless network.

1628.5 --> 1631.7: So all the nodes need to have access to the solution.

1632.5 --> 1634.0: It's a good question though.

1634.6 --> 1641.6: I think that's something that we'll have to navigate in terms of perhaps like with the ISO communities.

1642.2 --> 1649.4: A lot of the time they have this like year where the information is disclosed, but only within the working group.

1649.4 --> 1654.9: And then once it becomes a standard, it obviously gets distributed to everyone in an open source manner.

1655.5 --> 1658.7: So we could consider that and it's some good feedback.

1659.1 --> 1665.5: I don't know if you've got any comments on that from the Zoom side of things.

1666.4 --> 1669.4: Thank you very much for that comment.

1669.8 --> 1675.6: So the result of that competition should be public available.

1675.6 --> 1690.3: And in general, when NIST holds this kind of competition, all applicants need to give up their intellectual property to make it public available.

1691.5 --> 1695.3: So we need to discuss about that IPR policy for this competition,

1695.4 --> 1704.3: but the migration method decided throughout this competition should be public available, free of charge.

1706.2 --> 1721.0: Yeah, so just to replay that, the method and the IP behind the method could be made, I guess, private but disclosed within the competition.

1721.4 --> 1731.2: But if you are a winner and chosen as the migration, that has to be open source because all the nodes need to know the PQC.

1739.6 --> 1742.1: Cool. Do we have any other questions?

1742.7 --> 1743.7: Oh, yeah, go ahead.

1745.9 --> 1752.8: I worked for a number of years with the quantum scientists at UNSW and I really love this area.

1753.9 --> 1756.8: How are we going to test the solutions?

1757.2 --> 1763.9: Is there a live sort of environment in which you can actu

ally see if it works?

1763.9 --> 1769.1: How is the testing intended to have the new competition?

1769.9 --> 1777.3: Yeah, so at begin, we're looking to be part of the neutral body that creates the test bed.

1777.9 --> 1785.1: So we described we'll spin up nodes with global presence.

1785.5 --> 1792.3: So definitely some in North America, some in Japan, some in Australia, some in London, at least, at minimum.

1792.3 --> 1806.5: And we need to come up with the criteria for what is sufficient decentralization for that test to be relevant, that test bed to be relevant and have the right decentralization, but also information.

1809.4 --> 1813.0: And it will be facilitated through this neutral body.

1813.0 --> 1821.6: So I'm happy to speak about that further and have some of your quantum researchers be a part of that process.

1821.8 --> 1829.0: And perhaps we could set up the Australian nodes through sort of your network.

1829.1 --> 1829.9: That would be brilliant.

1839.9 --> 1841.8: There we go.

1842.0 --> 1857.7: So one question, do you see a branching in the way that the post-quantum migration will carry on between Bitcoin and Ethereum?

1857.7 --> 1870.3: I see a lot of sense in looking at retrofitting post-quantum cryptography into the Ethereum ecosystem, because there's a lot of embedded logic that's built there.

1870.8 --> 1876.1: But there's much less of such embedded logic in the Bitcoin ecosystem.

1876.1 --> 1891.4: Do you think that it's likely that you might see the Ethereum ecosystem move towards some form of migration of code versus the Bitcoin ecosystem moving to a migration of value?

1892.0 --> 1901.7: So in other words, I bridge into another already post-quantum chain rather than try and retrofit something into something that exists.

1904.5 --> 1905.9: I guess I can answer that.

1907.0 --> 1908.5: So thank you for asking.

1908.7 --> 1909.8: That's a great question.

1910.2 --> 1918.6: So actually, there are some chains saying we are quantum ready from the very beginning, from scratch, from the first block.

1918.6 --> 1922.6: And also they are targeting to support the EVM.

1923.0 --> 1939.7: So to place them as the quantum haven, in case the migration in Ethereum doesn't work well, maybe they can move their contract or asset to their brand new chain.

1939.7 --> 1943.2: And I think that's one of the form of the migration too.

1943.3 --> 1952.6: But that's also equivalent as the hard working and leaving all knowledge and community behind and moving on to the new one.

1953.0 --> 1956.5: And I don't really know that's the best way to go.

1956.9 --> 1962.1: And talking about the differentiation between Bitcoin and Ethereum.

1963.1 --> 1968.0: Well, yeah, I mean, talking about if we take smart contract as an example.

1968.7 --> 1973.1: Well, yeah, I mean, the migration of the smart contract also look hard.

1973.4 --> 1975.5: And as you say, it's kind of embedded already.

1977.2 --> 1989.1: But at the same time, the Ethereum migration is also kind

of having better progress looking from the outside.

1990.1 --> 1996.7: The Ethereum foundation is launching some initiative to support the PQC migration for Ethereum.

1997.5 --> 2005.1: And thanks for those embedded account abstraction technique or something like some flexibility.

2005.7 --> 2012.4: They can also have wider options to take for the post-quantum migration.

2012.4 --> 2022.4: So I think there's pros and cons between, you know, the Bitcoin simple value versus, you know, Ethereum embedded, more flexible code bases.

2023.8 --> 2028.6: And, well, I kind of get drunk a minute, but I think that's the answer to your question.

2034.1 --> 2035.3: Yes, I think that answer.

2041.0 --> 2042.0: Appreciate it.

2042.1 --> 2046.9: I have to leave this call too.

2048.2 --> 2049.1: There's a hard stop.

2052.6 --> 2055.6: What time this session?

2062.8 --> 2071.8: Yeah, I think that answers my question, but I think you'll see there's more.

2072.8 --> 2079.8: I suspect you will find that there's more impetus to migrate something that has greater value.